• Unit 3 • AOS 1 Data Analysis • Lesson 1

What kind of data is this?

1A Types of data · 1B Displaying and describing categorical data

APK | Learn | Worked examples | You try | Exit tasks

TODAY'S LESSON

Today's lesson

LEARNING INTENTIONS

- ☐ To be able to classify data as categorical or numerical.
- ☐ To be able to further classify categorical data as nominal or ordinal.
- ☐ To be able to further classify numerical data as discrete or continuous.
- ☐ To be able to construct frequency and percentage frequency tables for categorical data.
- ☐ To be able to construct bar charts and percentage bar charts from frequency tables.
- ☐ To be able to interpret and describe frequency tables and bar charts.
- ☐ To be able to answer statistical questions about a categorical variable.

LESSON SEQUENCE

1. APK · Year 11 skills and two VCAA questions
2. Learn 1A · types of data
3. Worked example 1 → you try
4. Learn 1B · frequency tables → WE 2 → you try
5. Bar charts · WE 3 → you try
6. Segmented bar charts · learn → WE 4 → you try
7. Describing the distribution · learn → WE 5 → you try
8. Exit tasks · three VCAA questions
9. Check-in and homework

SUCCESS CRITERIA

By the end of this lesson you'll be able to say...

- [ ] I can identify types of data.
- [ ] I can construct a frequency table for categorical data.
- [ ] I can construct a bar chart from a frequency table.
- [ ] I can construct a percentage segmented bar chart from a frequency table.
- [ ] I can describe the distribution of a categorical variable.

APK · FROM YEAR 11

What do you remember?

Classify the following data as nominal, ordinal, discrete or continuous:

a The time between people arriving at the coffee shop.

b The number of people in the queue at the coffee shop.

c Customer's coffee preference (latte, cappuccino, black).

d Customer's rating of the coffee (excellent, quite good, not that good).

a **Continuous**: time is measured.

b **Discrete**: people are counted.

c **Nominal**: the categories just name the coffee.

d **Ordinal**: the ratings have a natural order.

Quick write

APK · VCAA 2021 EXAM 1 Q1

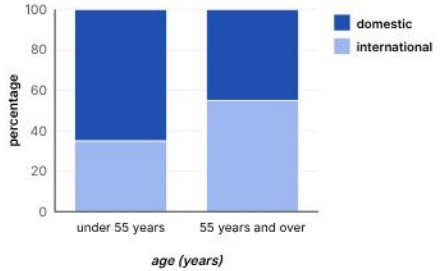
VCAA: **70%** of students answered correctly

A real VCAA question you can already do

The percentage segmented bar chart provided shows the *age* (under 55 years, 55 years and over) of visitors at a travel convention, segmented by *preferred travel destination* (domestic, international).

The variables *age* (under 55 years, 55 years and over) and *preferred travel destination* (domestic, international) are

A.	**both categorical variables.**	70%
B.	both numerical variables.	3%
C.	a numerical variable and a categorical variable respectively.	21%
D.	a categorical variable and a numerical variable respectively.	4%
E.	a discrete variable and a continuous variable respectively.	2%



SOLUTION · A

Age has been recorded in groups (ordinal) and destination is a name (nominal). Both are categorical.

WHAT'S THE DISTRACTOR?

DISTRACTOR · C

Students saw *age* and assumed it was numerical. Here it has been recorded in groups (under 55, 55 and over), so it is categorical.

Multiple choice

Cold calling

APK · VCAA 2020 EXAM 1 Q10

VCAA: **57%** of students answered correctly

And another

The data in the table shown was collected in a study of the association between the variables *frequency of nightmares* (low, high) and *snores* (no, yes).

The variables in this study, *frequency of nightmares* (low, high) and *snores* (no, yes), are

A.	**ordinal and nominal respectively.**	57%
B.	nominal and ordinal respectively.	5%
C.	both numerical.	3%
D.	both ordinal.	24%
E.	both nominal.	11%

frequency of nightmares	*snores*		total
	no	yes	
low	80	58	138
high	11	12	23
total	**91**	**70**	**161**

SOLUTION · A

Low and high have a natural order: **ordinal**. No and yes only name a group: **nominal**.

WHAT'S THE DISTRACTOR?

DISTRACTOR · D

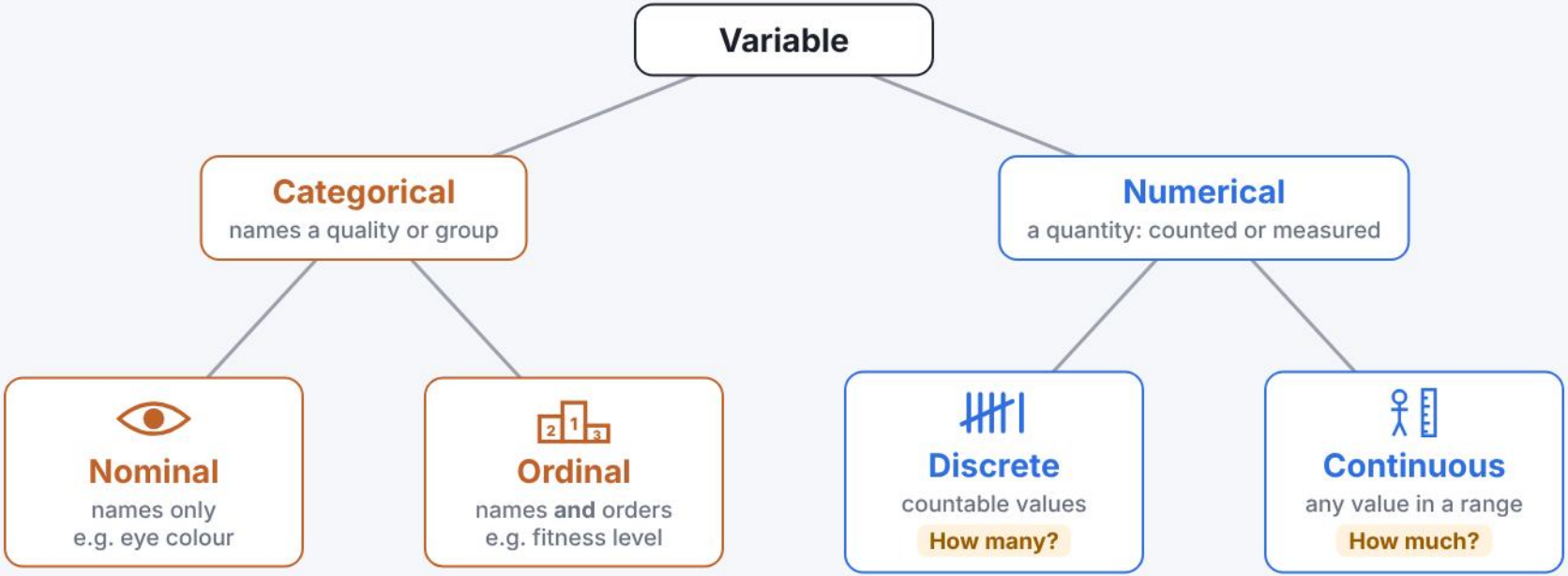
Students treated *snores* (no, yes) as ordered. No and yes only name a group, so it is nominal, not ordinal.

Multiple choice

Cold calling

LEARN · 1A

Two types of variable, each with two sub-types



LEARN · 1A

When you’re not sure: two tests

TEST 1 · CAN YOU AVERAGE IT?

It makes sense to calculate the average *weight* of a group of people, but not the average *house number* in a street.

Numbers that are just labels are **categorical**.

TEST 2 · HOW WAS IT RECORDED?

weight recorded in kilograms → **numerical**

weight recorded as ‘underweight’, ‘normal weight’, ‘overweight’ → **categorical**

The variable name alone doesn’t decide.

Class discussion

LEARN · 1A · THINK-PAIR-SHARE

A university survey: classify each variable

Decide with your partner. Each click reveals the next card.

Height
e.g. 173, 179, 167 (cm)

Numerical · continuous

Age
e.g. 18, 19, 22 (years)

Numerical · continuous*

Pulse rate
e.g. 86, 82, 96 (beats/min)

Numerical · discrete (counted beats)

Study mode
C = on-campus, O = online

Categorical · nominal

Fitness level
1 = high, 2 = medium, 3 = low

Categorical · ordinal

Weight
e.g. 57, 58, 62 (kg)

Numerical · continuous

* Age is continuous (time passes continuously), even when it's recorded in whole years.

Survey data: Cambridge General Mathematics Units 3&4, 1A



Think, pair, share

WORKED EXAMPLE 1 · 1A

Video

Types of data

Classify the following data as nominal, ordinal, discrete or continuous.

a The number of chocolate chips in each of 10 cookies is counted.

b The time taken for 20 students to complete a puzzle is recorded in seconds.

c Member of a football club were asked to rate how they felt about the current coach, 1 = Very satisfied, 2 = Satisfied, 3 = Indifferent, 4 = Dissatisfied, 5 = Very dissatisfied.

d Students are asked to each choose their preferred colour from the list 1 = Blue, 2 = Green, 3 = Red, 4 = Yellow.

e Students weights were classified as ‘less than 60kg’, ‘60kg - 80kg’ or ‘more than 80kg’.

a Discrete, as the number of chocolate chips will only take whole number values.

b Continuous, as the data can take any value, limited only by the accuracy to which the time can be measured.

c Ordinal, as the numbers in this data do not represent quantities, they represent each person’s level of approval of the coach.

d Nominal, as the numbers in this data again do not represent quantities, they represent colours.

e Ordinal, as each student’s weight has been recorded into three categories which can be ordered.

Cold calling

YOU TRY · 1A

Types of data

Classify the following variables as either nominal, ordinal, discrete or continuous.

a *car brand* (1 = Toyota, 2 = Holden, 3 = Ford, 4 = other)

b *number of employees*

c *weight of textbook* (kg)

d *height of basketball players* (cm)

e *perfume brand*

a Nominal: the numbers are labels, not quantities.

b Discrete: employees are counted.

c Continuous: weight is measured.

d Continuous: height is measured.

e Nominal: brands are names with no order.

Quick write

LEARN · 1B

From raw data to a frequency table

A **frequency table** lists each category and how often it occurs: as a number and as a percentage. Here: the *study mode* of 60 university students.

online	online	campus	online	online	online	campus	online	campus	campus
campus	online	campus	online	online	online	campus	campus	campus	online
campus	online	campus	online	campus	campus	campus	online	campus	online
campus	online	campus	online	online	online	campus	online	campus	online
campus	online	online	online	online	online	online	campus	campus	online
campus	online	online	online	campus	online	campus	campus	campus	online

study mode	Frequency	
	Number	%
campus	27	45.0
online	33	55.0
Total	**60**	**100.0**

PERCENTAGE FREQUENCY

$$\text{percentage frequency} = \frac{\text{number of times a value occurs}}{\text{total number of values}} \times 100\%$$

1. **Set up the table** — One row for each category.
2. **Count** — Count each category.
3. **Total** — $27 + 33 = 60$
4. **% campus** — $\frac{27}{60} \times 100\% = 45.0\%$
5. **% online** — $\frac{33}{60} \times 100\% = 55.0\%$
6. **Check** — $45.0 + 55.0 = 100\%$

Data: Cambridge General Mathematics Units 3&4, 1B

Class discussion

WORKED EXAMPLE 2 · 1B

▶ Video

Frequency table for a categorical variable

A group of 11 preschool children were asked to choose between chocolate and vanilla ice-cream (C = chocolate, V = vanilla):

C V V C C V C C C V V

Construct a frequency table (including percentage frequencies) to display the data.

flavour	Frequency	
	Number	Percentage
chocolate	6	54.5
vanilla	5	45.5
Total	**11**	**100.0**

1 Set up the table
One row per flavour, columns for number and percentage.

2 Count
chocolate 6, vanilla 5

3 Total
$6 + 5 = 11$

4 % chocolate
$\frac{6}{11} \times 100\% = 54.5\%$

5 % vanilla
$\frac{5}{11} \times 100\% = 45.5\%$

6 Check
$54.5 + 45.5 = 100\%$
If it totals 99.9 or 100.1, leave it: that's rounding, and it's fine.

Cold calling

YOU TRY · 1B

Frequency table for a categorical variable

The students in a prep class were asked the question, 'Would you describe your teacher's height as short, average or tall?'. Their responses were as follows:

average	short	tall	tall	short	short
average	average	tall	short	average	short
average	average	average	tall	tall	short
average	tall	tall	average	average	tall

Use this data to create a frequency table displaying both frequency and percentage frequency, correct to the nearest decimal place.

teacher's height	frequency	
	number	**%**
short	6	25.0
average	10	41.7
tall	8	33.3
total	**24**	**100.0**

1 Set up
One row per response.

2 Count
short 6, average 10, tall 8

3 Total
$6 + 10 + 8 = 24$

4 % short
$\frac{6}{24} \times 100\% = 25.0\%$

5 % average
$\frac{10}{24} \times 100\% = 41.7\%$

6 % tall
$\frac{8}{24} \times 100\% = 33.3\%$

7 Check
$25.0 + 41.7 + 33.3 = 100\%$
99.9 or 100.1 is fine: that's rounding.

Quick write

WORKED EXAMPLE 3 · 1B

▶ Video

Constructing a bar chart from a frequency table

The climate type of 23 countries is classified as ‘cold’, ‘mild’ or ‘hot’. The results are summarised in the table opposite.

a What is the level of measurement of the variable climate type?

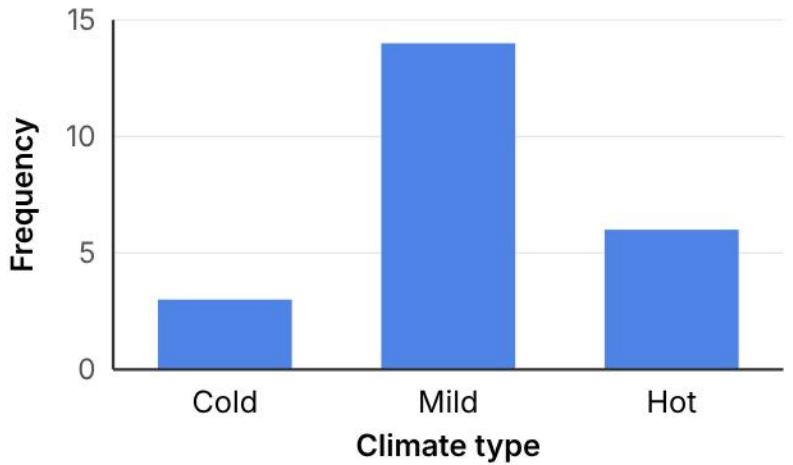
b Construct a frequency bar chart to display this information.

Climate type	Frequency	
	Number	Percentage
Cold	3	13.0
Mild	14	60.9
Hot	6	26.1
Total	**23**	**100.0**

✏ **Use a ruler** for axes and bars, and label both axes.

a Ordinal: the data both group the countries by climate type and put the groups in a natural order of ‘warmth’.

b Frequency on the vertical axis, one bar per category, gaps between bars.



Cold calling

YOU TRY · 1B

Constructing a bar chart from a frequency table

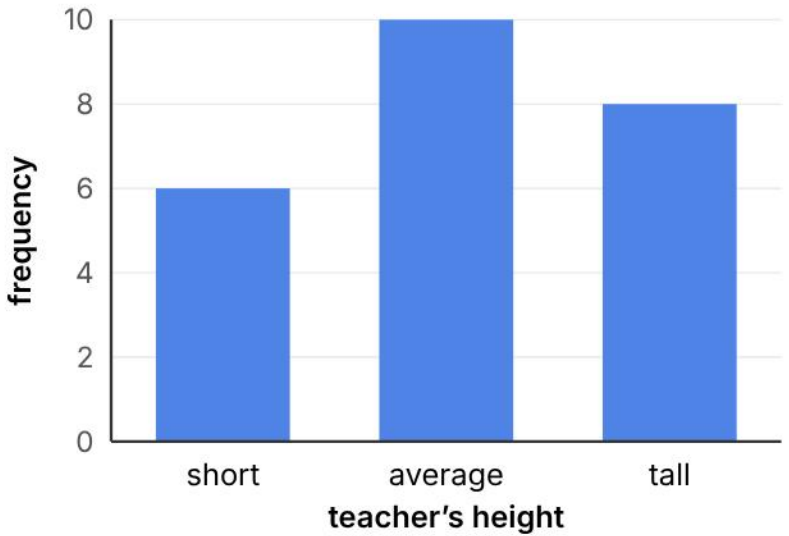
24 students in a prep class were asked the question, 'Would you describe your *teacher's height* as short, average or tall?'. Their responses are recorded in the frequency table shown.

Use the frequency table to construct a frequency bar chart.

teacher's height	frequency	
	number	%
short	6	25.0
average	10	41.7
tall	8	33.3
total	**24**	**100.0**

✏ **Use a ruler** for axes and bars, and label both axes.

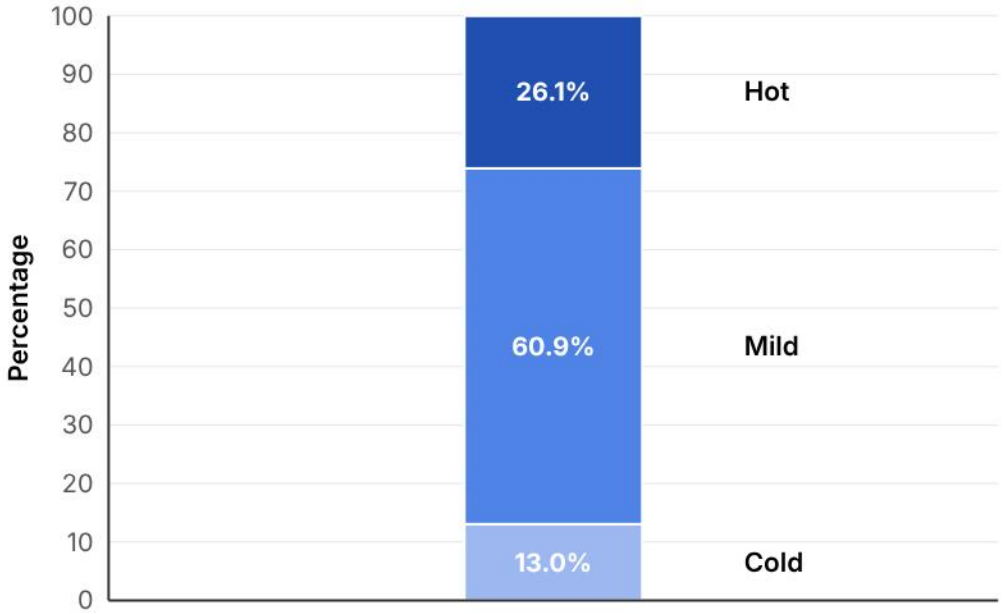
Frequency on the vertical axis (up to 10), one bar per category, gaps between bars.



Quick write

LEARN · 1B

From a bar chart to a segmented bar chart



1 · STACK THE BARS

Same frequencies, one on top of the other. The total height is 23.

2 · CONVERT TO PERCENTAGES

Each segment becomes its percentage, so the bar is exactly 100% tall.

WHY?

One compact bar shows each category's **share** of the whole, and makes groups easy to compare.

Data: Cambridge General Mathematics Units 3&4, Example 3

Class discussion

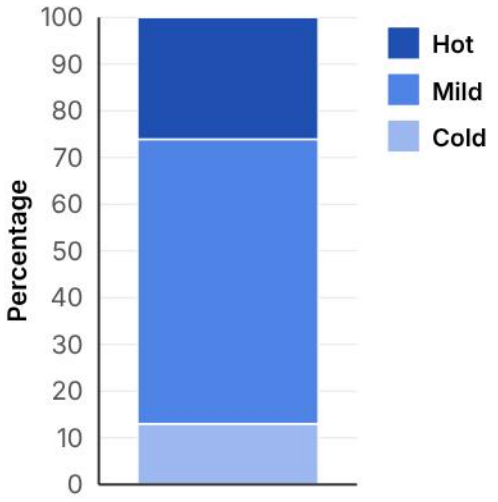
WORKED EXAMPLE 4 · 1B

▶ Video

Constructing a percentage segmented bar chart from a frequency table

The climate type of 23 countries is classified as ‘cold’, ‘mild’ or ‘hot’. Construct a percentage frequency segmented bar chart to display this information.

Climate type	Frequency	
	Number	Percentage
Cold	3	13.0
Mild	14	60.9
Hot	6	26.1
Total	**23**	**100.0**



1 One bar
Height 100%, no label on the horizontal axis.

2 Cold
Segment from 0 to 13%.

3 Mild
From 13% to $13 + 60.9 = 73.9\%$.

4 Hot
From 73.9% to 100%.

5 Legend
Shade each segment and add a legend.

Cold calling

YOU TRY · 1B

Constructing a percentage segmented bar chart from a frequency table

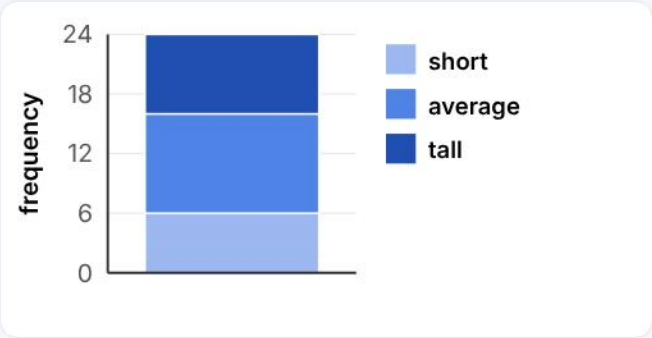
24 students in a prep class were asked the question, 'Would you describe your *teacher's height* as short, average or tall?'. Their responses are recorded in the frequency table shown.

a Use the frequency table to construct a segmented bar chart.

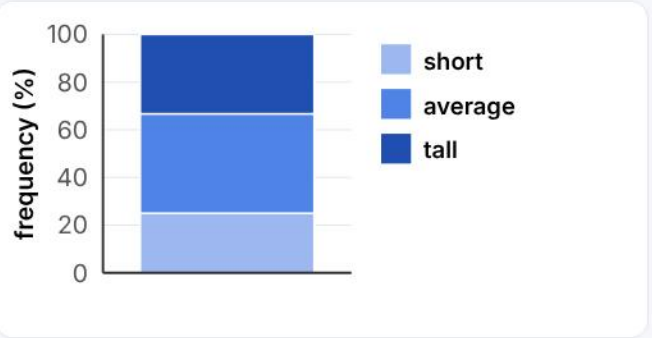
b Use the frequency table to construct a percentage segmented bar chart.

teacher's height	frequency	
	number	%
short	6	25.0
average	10	41.7
tall	8	33.3
total	**24**	**100.0**

a Segments end at 6, $6 + 10 = 16$ and $16 + 8 = 24$.



b Segments end at 25%, $25 + 41.7 = 66.7\%$ and 100%.



Quick write

LEARN · 1B

Describing the distribution of a categorical variable

THE MODE

The most frequently occurring category: the **tallest bar** or **longest segment**.

A REPORT INCLUDES

the context and how many

the modal category

percentages for the others

Context: 40 students were asked how they travel to school: car 22 (55%), bus 12 (30%), walk 6 (15%).

The **mode of travel to school** of **40** **students** were classified as being **'car', 'bus' or 'walk'**. The **majority** of the **students**, **55%**, were found to **travel by car**. Of the remaining **students**, **30%** were found to **catch the bus**, while **15%** were found to **walk**.

Use 'majority' only when the modal category is more than 50%. Otherwise, say 'the most common'.

Class discussion

WORKED EXAMPLE 5 · 1B

Video

Describing the distribution of a categorical variable

In an investigation of the variation of climate type across countries, the climate types of 23 countries were classified as ‘cold’, ‘mild’ or ‘hot’. The data are displayed in a frequency table to show the percentages.

Use the information in the frequency table to write a concise report on the distribution of climate types across these 23 countries.

Climate type	Frequency	
	Number	%
Cold	3	13.0
Mild	14	60.9
Hot	6	26.1
Total	**23**	**100.0**

Report: The climate types of 23 countries were classified as being, ‘cold’, ‘mild’ or ‘hot’. The majority of the countries, 60.9%, were found to have a mild climate. Of the remaining countries, 26.1% were found to have a hot climate, while 13.0% were found to have a cold climate.

Cold calling

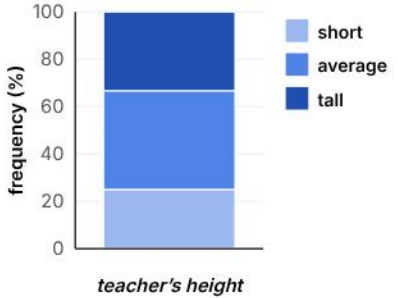
YOU TRY · 1B

Describing the distribution of a categorical variable

24 students in a prep class were asked the question, 'Would you describe your *teacher's height* as short, average or tall?'. Their responses are shown in the given percentage segmented bar chart.

a Find the modal category of the data set.

b Describe the distribution of the data set.



a **Average**: the longest segment (41.7%).

b

The **responses** of **24** **prep students** were classified as being **'short', 'average' or 'tall'** . The **most common response** of the **students** , **41.7%** , were found to **describe their teacher as average** . Of the remaining **students** , **33.3%** were found to **say tall** , while **25%** were found to **say short** .

Quick write

EXIT TASK 1 OF 3 · VCAA 2019 EXAM 2 Q1A

VCAA: **81%** scored 1/1

Show what you know

The table shows the *day number* and the *minimum temperature*, in degrees Celsius, for 15 consecutive days in May 2017.

day number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
***minimum temperature* (°C)**	12.7	11.8	10.7	9.0	6.0	7.0	4.1	4.8	9.2	6.7	7.5	8.0	8.6	9.8	7.7

Which of the two variables in this data set is an ordinal variable? (1 mark)

SOLUTION

day number. It labels each day and has a natural order, but it is not counted or measured, so it is categorical (ordinal). *minimum temperature* is measured, so it is numerical.

Common error: answering 'neither' because *day number* uses numbers.

Quick write

EXIT TASK 2 OF 3 · VCAA 2021 EXAM 2 Q1A

VCAA: **52%** scored 1/1

In the sport of heptathlon, athletes compete in seven events.
These events are the 100 m hurdles, high jump, shot-put, javelin, 200 m run, 800 m run and long jump.
Fifteen female athletes competed to qualify for the heptathlon at the Olympic Games.
Their results for three of the heptathlon events – high jump, shot-put and javelin – are shown in the table.

athlete number	*high jump* (metres)	*shot-put* (metres)	*javelin* (metres)
1	1.76	15.34	41.22
2	1.79	16.96	42.41
3	1.83	13.87	46.53
4	1.82	14.23	40.53
5	1.87	13.78	40.62
6	1.73	14.50	45.62
7	1.68	15.08	42.33
8	1.82	13.13	40.88
9	1.83	14.22	39.22
10	1.87	13.62	42.51
11	1.87	12.01	42.75
12	1.80	12.88	38.12
13	1.83	12.68	42.65
14	1.87	12.45	41.32
15	1.78	11.31	42.88

Write down the number of numerical variables in the table. (1 mark)

SOLUTION

3: *high jump*, *shot-put* and *javelin* are measured. *athlete number* only labels each athlete, so it is categorical.

Quick write

EXIT TASK 3 OF 3 · VCAA 2017 EXAM 1 Q5 (ADAPTED)

VCAA (original question): **76%** of students answered correctly

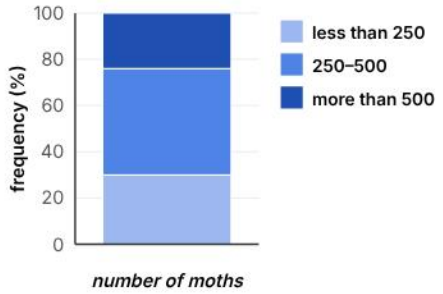
Show what you know

A study was conducted that investigated the *number of moths* caught in a sugar moth trap (less than 250, 250–500, more than 500). The results are summarised in the percentage segmented bar chart shown.

There were 300 sugar traps.

The number of sugar traps that caught less than 250 moths is closest to

A. 30	21%	**B.** **90**	76%	**C.** 250	2%
D. 300	1%	**E.** 500	1%		



SOLUTION · B

The 'less than 250' segment reaches 30%: $0.30 \times 300 = 90$

WHAT'S THE DISTRACTOR?

DISTRACTOR · A

Students read the 30% straight off the chart as 30 traps, without finding 30% of the 300 traps.

Multiple choice

Quick write

CHECK-IN

How did you go?

- ☐ I can identify types of data.
- ☐ I can construct a frequency table for categorical data.
- ☐ I can construct a bar chart from a frequency table.
- ☐ I can construct a percentage segmented bar chart from a frequency table.
- ☐ I can describe the distribution of a categorical variable.

Thumbs up, sideways or down for each one, then rate it on your worksheet: ✓ got it · ~ nearly · × not yet



True or false

HOMEWORK

Homework

CAMBRIDGE · EXERCISE 1A

Types of variables: categorical or numerical	Q1
Categorical variables: nominal or ordinal	Q2
Numerical variables: discrete or continuous	Q3
Exam 1 style questions	Q4–6

CAMBRIDGE · EXERCISE 1B

Constructing a frequency table for categorical data	Q1–3
Constructing a percentage segmented bar chart from a frequency table	Q4–5
Using the distribution of a categorical variable to answer statistical questions	Q6–8
Exam 1 style questions	Q9

REVISION BOOK

The **1A** and **1B** sections